

Refuting a Conjecture of Umans and Wang on Arithmetic-Progression Divisor Covers

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Abstract

An n -divisor set is a finite set of positive integers containing a multiple of every integer from 1 through n . Umans and Wang proposed, as the arithmetic-progression version of their Strong (α, β) -Divisor Conjecture, an n -divisor arithmetic progression having at most $n^{2\beta}$ terms, each of magnitude at most $\exp(n^\alpha)$. We prove unconditionally that an n -divisor arithmetic progression of height H with $\log H = o(\sqrt{n})$ must have length

$$L \geq \left(\sqrt{\frac{8}{27}} - o(1) \right) \frac{n^{3/4}}{\sqrt{\log n}}.$$

Consequently, the arithmetic-progression version is false whenever $\alpha < 1/2$ and $\beta < 3/8$. In particular, it is false at the proposed point $(\alpha, \beta) = (1/3, 1/3)$, even if both bounds are relaxed by $n^{o(1)}$ at the exponent level. The proof uses primes in a fixed band below $\sqrt{n}$ to turn semiprime divisibility into a finite incidence structure. An elementary bounded-degree linear-space estimate then gives the result. This theorem concerns the one-dimensional arithmetic-progression version only; it does not disprove the higher-rank Strong Divisor Conjecture.

1 Introduction and statement of results

For $n \in \mathbb{Z}_{>0}$, a finite set $A \subset \mathbb{Z}_{>0}$ has the n -divisor property if every $d \in \{1, \dots, n\}$ divides at least one member of A . Umans and Wang introduced conjectures asking for n -divisor sets with additional difference-set structure and small height [4]. Proposition 3.4 and Section 6 of their paper single out a formally stronger sufficient statement: for infinitely many n , there should be a positive arithmetic progression

$$A = \{b + ic : 0 \leq i < L\}, \quad L \leq n^{2\beta},$$

whose members have magnitude at most $\exp(n^\alpha)$ and which has the n -divisor property. They call this the *Arithmetic Progression Version* of the Strong (α, β) -Divisor Conjecture. It is an informally named version rather than a separately numbered conjecture.

Our main theorem gives a quantitative obstruction to every such progression of sub-square-root logarithmic height.

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Theorem 1.1 (Quantitative AP lower bound). *Let n tend to infinity through any unbounded sequence, and for each such n let*

$$A_n = \{b_n + ic_n : 0 \leq i < L_n\} \subset \mathbb{Z}_{>0}$$

be an arithmetic progression with $b_n, c_n \in \mathbb{Z}$ and with the n -divisor property. Put $H_n = \max A_n$. If

$$\log H_n = o(\sqrt{n}),$$

then

$$L_n \geq \left(\sqrt{\frac{8}{27}} - o(1) \right) \frac{n^{3/4}}{\sqrt{\log n}}. \quad (1)$$

Equivalently, for every $\varepsilon > 0$, inequality (1) holds with $\sqrt{8/27} - \varepsilon$ in place of $\sqrt{8/27} - o(1)$ for all sufficiently large n in the sequence.

Corollary 1.2 (Exponent exclusion). *Fix $\alpha, \beta \geq 0$ with*

$$\alpha < \frac{1}{2}, \quad \beta < \frac{3}{8}.$$

There is no unbounded sequence of integers n admitting positive n -divisor arithmetic progressions satisfying

$$L_n \leq n^{2\beta+o(1)}, \quad H_n \leq \exp(n^{\alpha+o(1)}).$$

In particular, for every sufficiently large n there is no such progression under the stronger literal bounds $L_n \leq n^{2\beta}$ and $H_n \leq \exp(n^\alpha)$.

Taking $\alpha = \beta = 1/3$ disproves the Arithmetic Progression Version at the one-third point, including its exponent-level relaxation. The proof is unconditional. Its only analytic input is the prime number theorem, used in the standard equivalent forms

$$\pi(x) \sim \frac{x}{\log x}, \quad \vartheta(x) := \sum_{p \leq x} \log p \sim x;$$

see, for example, [2, Chapter 6]. The finite-incidence estimate is proved below. It is related in spirit to classical finite-linear-space arguments such as those of de Bruijn and Erdős [1], but no external incidence theorem is required.

Discovery using Codex

The proof was discovered in an OpenAI Codex run using the `gpt-5.6-sol` model with reasoning effort set to `ultra`; see [3]. Codex also produced the initial write-up. Subsequent human review verified the proof, reviewed the citations, and revised the exposition. All named authors approve the final manuscript and take responsibility for its claims and any remaining errors.

Acknowledgments

The prompting strategy for the Codex discovery run borrowed several design ideas from the UCLA Moonshot Harness project [5].

Reproducibility

The mathematical argument is self-contained and does not rely on computer-assisted calculations or on access to the Codex transcript. The model identifier and reasoning setting of the discovery run are recorded in the preceding disclosure.

2 A bounded-degree linear-space lemma

By a *block* we mean an indexed proper subset $B_j \subsetneq V$ in an indexed family $\mathcal{B} = (B_j)_{j \in J}$. Blocks with distinct indices may be identical as subsets.

Lemma 2.1 (Bounded-degree linear cover). *Let V be a finite set of $v \geq 2$ points, and let $\mathcal{B} = (B_j)_{j \in J}$ be an indexed family of blocks. Suppose that*

1. *for every two distinct points $p, q \in V$, there exists a block B_j containing both p and q ;*
2. *blocks with distinct indices intersect in at most one point; and*
3. *every point lies in at most Δ blocks.*

Then

$$v \leq \Delta(\Delta - 1) + 1. \quad (2)$$

Proof. First fix a block B_j . Since it is proper, choose $u \in V \setminus B_j$. For every $q \in B_j$, the pair-covering assumption supplies a block containing u and q . The blocks obtained for distinct q have distinct indices: otherwise one of them would meet B_j in at least two points. All these blocks contain u , so $|B_j| \leq \Delta$.

Now fix $u \in V$. The at most Δ blocks containing u together contain every other point of V . Each such block has at most $\Delta - 1$ points other than u . Therefore

$$v - 1 \leq \Delta(\Delta - 1),$$

which is (2). □

3 Primes just below the square-root scale

We first prove a lower bound associated with any fixed prime band. The constant in Theorem 1.1 will follow by optimizing the band.

Lemma 3.1 (A fixed-band lower bound). *Under the hypotheses of Theorem 1.1, fix real constants $0 < a < b \leq 1$. Then*

$$L_n \geq \left(a\sqrt{2(b-a)} - o(1) \right) \frac{n^{3/4}}{\sqrt{\log n}}. \quad (3)$$

Proof. Write $x = \sqrt{n}$, and suppress the subscript n . A decreasing nonconstant progression can be reversed, so it may be written

$$A = \{u + ic : 0 \leq i < L\}, \quad u, c \in \mathbb{Z}_{>0}, \quad u + (L-1)c = H.$$

If the progression has only one distinct term, then the n -divisor property implies

$$\text{lcm}(1, \dots, n) \mid H.$$

The prime number theorem gives $\log \text{lcm}(1, \dots, n) = (1 + o(1))n$, contradicting $\log H = o(\sqrt{n})$. We may therefore assume $L \geq 2$ and $c \geq 1$. In particular, $c \leq H$.

Let

$$\mathcal{P}_0 = \{p \text{ prime} : ax \leq p \leq bx\}, \quad \mathcal{P} = \{p \in \mathcal{P}_0 : p \nmid c\}.$$

By the prime number theorem,

$$|\mathcal{P}_0| = (b - a + o(1)) \frac{x}{\log x}, \quad (4)$$

$$\sum_{p \in \mathcal{P}_0} \log p = (b - a + o(1)) x. \quad (5)$$

The number of primes in $\mathcal{P}_0$ dividing c is at most

$$\frac{\log c}{\log(ax)} \leq \frac{\log H}{\log(ax)} = o\left(\frac{x}{\log x}\right),$$

and

$$\sum_{\substack{p \in \mathcal{P}_0 \\ p|c}} \log p = \log \left(\prod_{\substack{p \in \mathcal{P}_0 \\ p|c}} p \right) \leq \log c \leq \log H = o(x).$$

Consequently, if $v = |\mathcal{P}|$, then

$$v = (b - a + o(1)) \frac{x}{\log x}, \quad \sum_{p \in \mathcal{P}} \log p = (b - a + o(1)) x. \quad (6)$$

If $L > a^2 n$, then (3) is immediate. We henceforth assume $L \leq a^2 n$. For every progression index i , define

$$B_i := \{p \in \mathcal{P} : p \mid u + ic\}.$$

We check the hypotheses of Lemma 2.1.

First, every pair of distinct primes $p, q \in \mathcal{P}$ lies in some block: indeed,

$$pq \leq b^2 n \leq n,$$

so the n -divisor property supplies a progression term divisible by pq .

Second, two differently indexed blocks meet in at most one point. If distinct p, q belonged to both B_i and B_j , with $i \neq j$, then

$$pq \mid (i - j)c.$$

Since $p, q \nmid c$, this implies $pq \mid i - j$. But

$$0 < |i - j| < L \leq a^2 n \leq pq,$$

a contradiction.

Third, every block is proper. If some B_i were all of $\mathcal{P}$, the product of the distinct primes in $\mathcal{P}$ would divide $u + ic$. By (6), this would imply

$$\log(u + ic) \geq \sum_{p \in \mathcal{P}} \log p = (b - a + o(1)) x,$$

contrary to $\log(u + ic) \leq \log H = o(x)$.

Finally, let Δ be the actual maximum point degree. For $p \in \mathcal{P}$, the congruence

$$u + ic \equiv 0 \pmod{p}$$

selects one residue class of indices modulo p , because $p \nmid c$. Therefore

$$\Delta \leq 1 + \frac{L}{ax}. \quad (7)$$

Put $t = L/(ax)$. Lemma 2.1 and (7) give

$$v \leq \Delta(\Delta - 1) + 1 \leq t^2 + t + 1.$$

Since $v \rightarrow \infty$, this forces

$$t \geq \frac{\sqrt{4v - 3} - 1}{2} = (1 - o(1))\sqrt{v}.$$

Using (6) and $\log x = \frac{1}{2} \log n$, we obtain

$$\begin{aligned} L &\geq (a - o(1))x \sqrt{\frac{(b - a)x}{\log x}} \\ &= \left(a\sqrt{2(b - a)} - o(1)\right) \frac{n^{3/4}}{\sqrt{\log n}}, \end{aligned}$$

which proves (3). □

Proof of Theorem 1.1. Take $b = 1$ and $a = 2/3$ in Lemma 3.1. The coefficient in (3) is

$$a\sqrt{2(b - a)} = \sqrt{\frac{8}{27}}.$$

The use of the endpoint $b = 1$ is legitimate: for distinct primes $p, q \leq \sqrt{n}$, one still has $pq \leq n$, exactly the range covered by the n -divisor property. The lemma now gives (1) directly. □

4 Consequences and scope

Proof of Corollary 1.2. Suppose, to the contrary, that qualifying progressions exist along an unbounded sequence of n . Since $\alpha < 1/2$,

$$\log H_n \leq n^{\alpha + o(1)} = o(\sqrt{n}),$$

so Theorem 1.1 applies. On the other hand, $\beta < 3/8$ gives

$$n^{2\beta + o(1)} = o\left(\frac{n^{3/4}}{\sqrt{\log n}}\right),$$

contradicting the theorem's lower bound for L_n . □

At the one-third point this gives the promised unconditional conclusion: there is no unbounded sequence of positive n -divisor arithmetic progressions with

$$L_n \leq n^{2/3+o(1)}, \quad H_n \leq \exp(n^{1/3+o(1)}).$$

Natural floor or ceiling choices in the number of terms are absorbed by the $o(1)$. Indexing the progression from 0 or from 1 merely changes its initial term.

Remark 4.1 (Positivity and zero). The n -divisor property in Definition 3.1 of Umans and Wang is explicitly defined for sets of positive integers. Thus zero is not an admissible progression term or divisor witness here. Allowing zero as a witness would make the condition vacuous, since every positive integer divides zero. A constant progression reduces to the singleton case treated in the proof.

Remark 4.2 (What has and has not been disproved). Proposition 3.4 of Umans and Wang [4] shows that the Arithmetic Progression Version implies the Strong (α, β) -Divisor Conjecture; no converse is asserted. The result above consequently does *not* disprove the full Strong $(1/3, 1/3)$ -Divisor Conjecture. A difference of two higher-rank generalized arithmetic progressions need not be a one-dimensional progression. The decisive step above uses

$$(u + ic) - (u + jc) = (i - j)c.$$

For a rank-two progression, the corresponding difference contains two independent coefficients, and the at-most-one-intersection argument does not follow.

Remark 4.3 (Endpoints). Theorem 1.1 does not cover the case $\log H = O(\sqrt{n})$, and Corollary 1.2 makes no claim when $\beta = 3/8$. Indeed, the lower bound $L \gg n^{3/4}/\sqrt{\log n}$ does not exclude $L \leq n^{3/4}$. The constant $\sqrt{8/27}$ is the outcome of the fixed-band argument and is not claimed to be optimal.

References

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